

Carefully formulate the negations of each of the statements in 5-7. Then prove each statement by contradiction.

**4.7.4.**     There is no least positive rational number.

Proof:

The given statement claims that  
there is no least positive rational  
number.

This statement is difficult to prove,  
since we have nothing to work with.

But we can negate the statement, and then  
prove by contradiction.

Suppose not. Suppose there is a least  
positive rational number.

Let  $r, R \in \mathbb{Q}^+$ , where  $r$  is the least  
positive rational number.

We must show that there is a positive  
rational number  $R$ , such that  $0 < R < r$ .

Let  $r = k/i$ , for some  $k, i \in \mathbb{Z}$ ,  $i \neq 0$ .  
Let  $R = r/2$

Therefore,  $0 < R < r$ , and this is what  
was to be shown by contradiction.

□

**Solution:**   There is no least positive rational number.